

This prompt extends and replaces all previous extension prompts. Merge all previous specifications into one unified master specification. Never create separate documents or alternative architectures.

# AI HOME DESIGNER

# MASTER PROMPT

# VERSION 2.0

This is the latest version of the master specification.

It supersedes all previous extension prompts.

Do NOT restart the project.

Do NOT redesign the architecture.

Do NOT remove or replace any existing modules.

Instead, merge this specification into the existing master documentation and extend it while maintaining full backward compatibility.

Everything generated previously remains valid unless explicitly extended in this document.

Treat this as the new canonical version of the AI Home Designer Master Specification.

MASTER PROMPT – COMPLETE AEC ECOSYSTEM, MARKETPLACE & MANUFACTURER PLATFORM

You are continuing the existing AI Home Designer Master Specification.

Do NOT redesign the platform.

Do NOT simplify anything.

Preserve every previous architectural decision.

Your task is to expand the platform into a complete Architecture, Engineering, Construction, Manufacturing and Building Commerce Ecosystem.

The objective is to create one integrated platform that covers the complete lifecycle of a building while connecting architects, engineers, contractors, manufacturers, suppliers and customers.

The result must be implementation-grade documentation suitable for enterprise software development.

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PRIMARY GOAL

The platform must support the entire ecosystem:

Idea



AI Design



Engineering



Technical Documentation



Permit Preparation



Construction Planning



Construction Execution



Quality Assurance



Building Operation



Renovation



Digital Twin



Marketplace

↓

Manufacturers

↓

Facility Management

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CREATE THE FOLLOWING MODULES

1. Complete Engineering Suite

Automatically design and validate

Structural Engineering

Mechanical Engineering

Electrical Engineering

Plumbing

HVAC

Fire Protection

Ventilation

Drainage

Water Supply

Rainwater Systems

Smart Home Systems

Solar Systems

Heat Pumps

Geothermal Systems

Battery Storage

EV Chargers

Security Systems

Lighting Design

Network Infrastructure

Generate complete engineering documentation.

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## 2. Manufacturer Platform

Create a dedicated platform for manufacturers.

Manufacturers can register.

After verification they may publish products.

Every product MUST contain complete engineering information.

Mandatory fields include:

Manufacturer

Brand

Product Name

Product Code

SKU

GTIN

Country

Certificates

Fire Rating

Pressure Strength

Compression Strength

Thermal Conductivity

Acoustic Rating

Density

Weight

Dimensions

Length

Width

Height

Available Sizes

Available Colors

Material Composition

Installation Instructions

Maintenance Instructions

Warranty

Life Expectancy

Environmental Product Declaration (EPD)

CE Certification

Declaration of Performance (DoP)

Technical Data Sheet (PDF)

Safety Data Sheet

BIM Object

CAD Files

IFC

DWG

DXF

3D Models

Photos

Videos

Pricing

Currency

Availability

Regional Availability

Distributor Network

Replacement Products

Compatible Products

Accessories

Required Installation Components

Recommended Products

Every category must define its own mandatory engineering attributes.

Examples:

Bricks

AAC Blocks

Concrete

Steel

Insulation

Windows

Doors

Roof Tiles

Wood

Pipes

Electrical Components

HVAC Equipment

Heat Pumps

Solar Panels

Fasteners

Paint

Adhesives

Flooring

Tiles

Furniture

Kitchen Systems

Bathroom Equipment

Every product must pass validation before publication.

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### 3. Supplier Marketplace

Verified suppliers can

Sell products

Publish prices

Manage stock

Offer promotions

Manage logistics

Provide delivery estimates

Support multiple warehouses

Support regional pricing

Support APIs

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#### 4. Intelligent Product Selection

During design the AI automatically selects products.

Selection based on

Building regulations

Structural requirements

Thermal performance

Fire requirements

Acoustic requirements

Budget

Availability

Distance to supplier

Delivery time

User preferences

Manufacturer preferences

Compatibility

Energy performance

Lifecycle cost

CO<sub>2</sub> footprint

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#### 5. AI Material Replacement

Automatically suggest

Cheaper alternatives

Premium alternatives

Locally available products

Products with better insulation

Products with higher fire resistance

Products with lower CO<sub>2</sub> emissions

Compatible replacements

Generate comparison reports.

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## 6. Manufacturer APIs

Manufacturers can synchronize

Product Catalog

Prices

Stock

Certificates

Documentation

Technical Specifications

Images

BIM Objects

Version History

Through REST APIs.

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## 7. BIM Product Library

Every product automatically generates

2D Symbol

3D Model

BIM Object

Material Properties

Performance Data

Installation Rules

Maintenance Schedule

Replacement Cycle

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## 8. Construction Marketplace

Architects

Engineers

Contractors

Subcontractors

Installers

Inspectors

Surveyors

Manufacturers

Suppliers

can create verified business profiles.

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## 9. AI Procurement

Automatically create

Purchase Orders

Material Lists

Supplier Comparison

Cost Optimization

Delivery Planning

Warehouse Planning

Waste Reduction

Cash Flow Planning

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## 10. Quality Verification

During construction

AI compares

Specified Product

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Delivered Product

↓

Installed Product

Verify

Certificates

Dimensions

Batch Numbers

Compliance

Expiry

Installation Quality

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## 11. Construction ERP

Support

Inventory

Warehouse

Orders

Invoices

Projects

Employees

Equipment

Vehicles

Maintenance

Supplier Management

Customer Management

Financial Dashboards

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## 12. Product Knowledge Base

Every product becomes part of a searchable engineering database.

Include

Standards

Certificates

Installation Guides

Engineering Rules

Compatibility Matrix

Performance History

Revision History

## Known Issues

AI learns from these specifications.

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### 13. Universal Product Validation

The system must reject incomplete products.

Every category defines mandatory engineering parameters.

A brick without dimensions, strength class or fire classification cannot be published.

An insulation product without lambda value cannot be published.

A heat pump without COP/SCOP data cannot be published.

No incomplete engineering products are allowed.

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### 14. Future Ecosystem

Design support for

Government integrations

National construction databases

Digital product passports

Carbon accounting

Circular economy

Recycling databases

Smart logistics

AI purchasing assistant

Automatic supplier negotiation

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## OUTPUT RULES

Generate implementation-grade documentation.

Never summarize.

Never use placeholders.

Generate database schemas, APIs, workflows, validation rules, user roles, permissions and algorithms.

Every module must integrate seamlessly with the existing AI Home Designer platform.

The result should define a complete AI-powered Architecture, Engineering, Construction and Manufacturing Ecosystem capable of becoming a global industry platform.